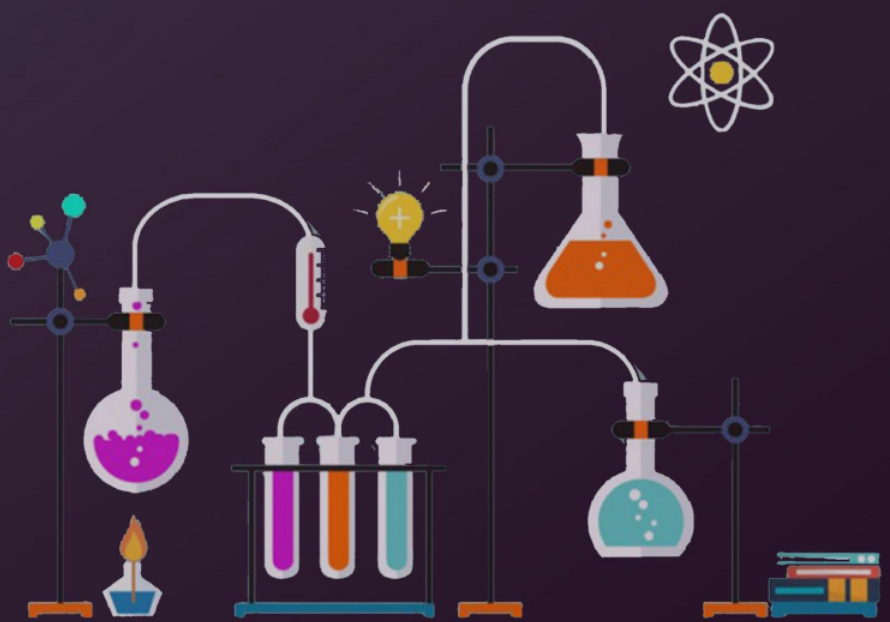
CHEMISTRY

Chemical Bonding



SHEET

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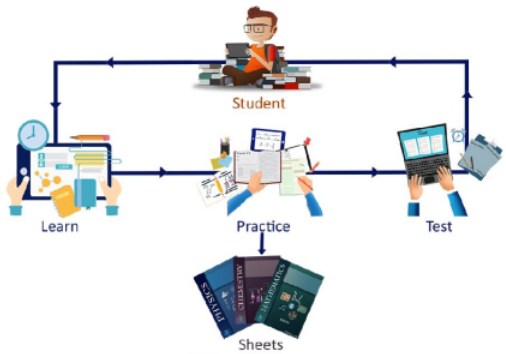
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Introduction	1			
2	Energy Profiles of Molecules & Lewis Theory				
3	Introduction to Ionic Bonding	2-5	1, 2, 27		6
4	Properties of Ionic Compound & Solubility	6-11	3	42, 43, 46	9
5	Fajans Rules	12-15	4-6		
6	Application of Fajans Rules	16-18	7, 28	30, 44, 45	
7	Covalent Bond Intro, Covalency	19-21			
8	Lewis Theory	22-28	8, 9	25	1, 5
9	VSEPR	29-39	10, 11, 12, 40, 41, 47	2, 18, 26, 27, 33	2, 11
10	VBT	40-47	13	1, 3, 4, 20	
11	Hybridisation	48-61	14, 26, 30, 31, 32, 39, 48	5, 24	7, 17
12	Bond Parameters	62-65	36-38	6-9, 39	18, 22
13	Hybridisation in Solid State, Bent and Drago's Rule	66-70	15	41	12, 23-25
14	Resonance, Back Bonding, Banana Bonding	71-74	16-18, 20, 21, 44, 45	10-13, 34-37	4
15	Molecular Orbital Theory	75-88	19, 22-24, 29, 33-35, 43, 49, 50	14, 28, 31, 32, 40	14, 15, 19
16	Dipole Moment	89-99	25, 46	16, 17, 19, 21-23, 38	8, 10,
17	Weak Bonds and Hydrogen Bonding	100-106	42	15, 29	3, 13, 16, 20, 21, 26

Exercise - 1 Concept Builder

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction

Q1. When two atoms combine to form a molecule :-
(A) Energy is released
(B) Energy is absorbed
(C) Energy is neither released nor absorbed
(D) Energy may either released o absorbed

Introduction to Ionic Bond and Solubility

Q2. An electrovalent bond or ionic bond is formed between ?
(A) Two electronegative atoms
(B) Two metals
(C) Electropositive and electronegative atoms
(D) Two electropositive atoms

Q3. An atom with atomic number 20 is most likely to combine chemically with the atom whose atomic number is ?
(A) 11 (B) 16
(C) 18 (D) 10

Q4. Bismuth does not form stable pentahalide because of ?
(A) Its higher electronegativity
(B) Its smaller size
(C) Inert pair effect
(D) Non availability of d-orbitals

Q5. An ionic compound $A^+ B^-$ is most likely to be formed when ?
(A) Ionization energy of A is low
(B) Electron affinity of B is high
(C) Electron affinity of B is low
(D) Both (A) and (B)

Properties of Ionic Compound & Solubility

Q6. Which of the following is not a correct statement about an ionic compound ?
(A) Higher the lattice energy, higher is melting point
(B) Higher the dipole moment of solvent, more the solubility
(C) Higher the lattice energy, more the solubility
(D) More difference in electronegativity, more is the ionic nature

Q7. Electrovalent compounds or ionic compounds do not show stereoisomerism. The reason is ?
(A) Presence of ions
(B) Strong electro static force of attraction
(C) Brittleness
(D) Non - directional nature of ionic bond

Q8. Capacity of a solvent to neutralise charge on an ionic compound is given by ?
(A) Solvation energy
(B) Dielectric constant
(C) Dipole moment
(D) Solubility

Q9. In which of the following solvents, KI has highest solubility ? The dielectric constant (ϵ) of each liquid is given in parentheses :-
(A) C_6H_6 (ϵ=0)
(B) $(CH_3)_2CO$(ϵ= 21)
(C) CH_3OH (ϵ=32)
(D) CCl_4 (ϵ=0)

Q10. Solid NaCl is a bad conductor of electricity because ?
(A) In solid NaCl there are no ions
(B) Solid NaCl is covalent
(C) In solid NaCl there is no mobility of ions
(D) In solid NaCl there are no electrons

Q11. Select correct order out of given options :
(A) $Na_2CO_3 > BaCO_3$: Lattice Energy
(B) BeO > SrO : lattice energy
(C) $Be^{2+} < Li^+$: Hydration energy
(D) $Be^{2+}(aq.) > Li^+(aq.)$: Ionic mobility

Fajans Rule

Q12. Which pair in the following has maximum and minimum ionic character respectively?
(A) LiCl, RbCl
(B) RbCl, $BeCl_2$
(C) $BeCl_2$, RbCl
(D) AgCl, RbCl

Q13. According to Fajans rules favorable condition to form covalent bond is ?
(A) small cation and large anion
(B) small cation and small anion
(C) large cation and large anion
(D) large cation and small anion

Q14. The polarizibility of the following compounds is ?
(A) $I^- > Br^- > Cl^- > F^-$
(B) $I^- > Br^- > F^- > C1^-$
(C) $I^- < Br^- < Cl^- < F^-$
(D) $I^- < Br^- < F^- < Cl^-$

Q15. Correct order of polarising power is :-
(A) $Cs^+ < K^+ < Mg^{2+} < Al^{3+}$
(B) $Al^{3+} < Mg^{2+} < K^+ < Cs^+$
(C) $Mg^+ < Al^{3+} < K^+ < Cs^+$
(D) $K^+ < Cs^+ < Mg^{2+} < Al^{3+}$

Application of Fajans Rule

Q16. Increasing order of thermal stability of -
I. K_2CO_3
II. $MgCO_3$
III. Na_2CO_3
(A) I < II < III (B) II < III < I
(C) II < I < III (D) I < III < II

Q17. Choose the compound from each of the following pairs that has more solubility in water.
(a) $BeSO_4$, $BaSO_4$
(b) NaCl, $MgCl_2$
(c) AgCl, AgI
(A) $BeSO_4$, $MgCl_2$, AgI
(B) $BeSO_4$,NaCl, AgCl
(C) $BaSO_4$, NaCl, AgCl
(D) $BaSO_4$, $MgCl_2$, AgI

Q18. Highest melting point would be of?
(A) $AlCl_3$
(B) LiCl
(C) NaCl
(D) $BeCl_2$

Covalent Bond Intro, Covalency

Q19. In Co-ordinate bond, the acceptor atoms must essentially contain in its valence shell an orbital:
(A) With paired electron
(B) With single electron
(C) With no electron
(D) With three electron

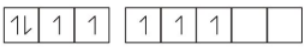
Q20. The bonds present in N_2O_5 are ?
(A) Only ionic
(B) Covalent & coordinate
(C) Only covalent
(D) Covalent & ionic

Q21. The maximum covalency is equal to ?
(A) the number of unpaired p-electrons
(B) the number of paired d-electrons
(C) the number of unpaired s and p-electrons
(D) total number of unpaired electron in ground state or in excited state

Lewis Theory

Q22. Which of the following configuration shows second excitation state of Iodine:-

(A) 

(B) 

(C) 

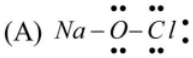
(D) 

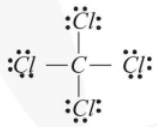
Q23. Which of the following fluorides does not exist ?

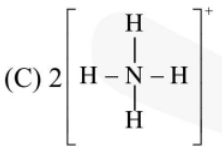
(A) NF_5 (B) PF_5

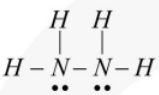
(C) AsF_5 (D) SbF_5

Q24. Which of the following Lewis dot diagrams is incorrect ?

(A) 

(B) 

(C) 

(D) 

Q25. NH_3 and BF_3 combine readily because of the formation of ?

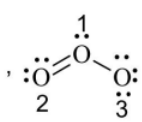
(A) a covalent bond

(B) a hydrogen bond

(C) a coordinate bond

(D) an ionic bond

Q26. It has been seen that in O_3, the central O atom is bonded to two other oxygen atoms as



Arrange oxygen atoms marked as 1, 2 and 3 in order of their decreasing formal charge:

(A) 1 < 2 < 3 (B) 1 > 3 > 2

(C) 1 > 2 > 3 (D) 2 > 1 > 3

Q27. Carbon has a Covalency of 2 in CO and 4 in CO_2 and CH_4, Its valency in C_2H_2 is :

(A) 3 (B) 4

(C) 2 (D) 1

Q28. Which is not an exception to octet rule ?

(A) BF_3 (B) $SnCl_4$

(C) BeI_2 (D) ClO_2

VSEPR

Q29. Which following set of compound/ion has linear geometry ?

(A) CH_4, NH_4^+, BH_4^-

(B) CO_3^{2-}, NO_4^-, BF_3

(C) NO_2^+, CO_2, N_3^-

(D) $BeCl_2$, BCl_3, CH_4

Q30. The species $[I(CN)_2]^-$ is ?

(A) Linear (B) Bent

(C) Pyramidal (D) none of these

Q31. Which of the following has a square planar structure ?

(A) NH_4^+ (B) BF_4^-

(C) XeF_4 (D) CCl_4

Q32. A σ bonded molecule MX_3 is T–shaped. The number of non–bonding pairs of electrons is ?

(A) 0

(B) 2

(C) 1

(D) Can be predicted only if atomic number of M is known.

Q33. Select the correct matching :

List I	List II
A : XeF_4	1. Pyramidal
B : XeF_6	2. T-shape
C : XeO_3	3. Distorted octahedral
D : $XeOF_2$	4. Square planar

	A	B	C	D
(A)	4	3	1	2
(B)	1	2	3	4
(C)	2	1	3	4
(D)	4	1	3	2

Q34. In which following set of compound/ion has linear shape

(A) CH_4, NH_4^+, BH_4^- (B) CO_3^{-2}, NO_4^-, BF_3

(C) NO_2^+, CO_2, XeF_2 (D)$BeCl_2$, BCl_3, CH_4

Q35. Which of the following set is not correct:–

(A) SO_3, O_3, NH_4^+ all have coordinate bonds

(B) H_2O, NO_2, ClO_2^-, all are 'V' shape molecules

(C) I_3–, ICl_2^-, NO_2^+ ; all are linear molecules

(D) SF_4, SiF_4 XeF_4 are tetrahedral in shape

Q36. Shape of a molecule having 4 bond pairs and two lone pairs of electrons, will be :-

(A) Square planar (B) Tetra hedral

(C) Linear (D) Octa hedral

Q37. Hybridisation in $XeOF_2$, XeO_2F_2 is sp^3d. But shape will be respectively:-

(A) T, 'V' shape

(B) T shape, (See-Saw)

(C) Both have T shape

(D) T shape, irregular octahedral

Q38. The shape of IF_4^+ will be :-

(A) Square planar

(B) Tetranedral

(C) Pentagonal bipyramidal

(D) Distorted tetrahedral

Q39. Amongst CO_3^{2-}, AsO_3^{3-}, XeO_3, ClO_3^-, BO_3^{3-} and SO_3^{2-} the non-planar species are :-

(A) XeO_3 , ClO_3^- , SO_3^{2-},AsO_3^{3-}

(B) AsO_3^{3-} , XeO_3 , CO_3^{-2}

(C) BO_3^{3-} , CO_3^{2-} , SO_3^{2-}

(D) AsO_3^{3-} , BO_3^{3-} , CO_3^{2-}

VBT

Q40. A sigma bond is formed by the overlapping of :–

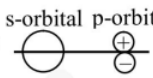
(A) s-s orbital alone

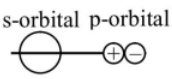
(B) s and p orbitals alone

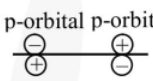
(C) s–s, s–p or p–p orbitals along internuclear axis

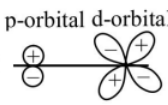
(D) p–p orbital along the sides

Q41. Which of the following leads to bonding ?

(A) 

(B) 

(C) 

(D) 

Q42. The strength of bonds by 2s - 2s, 2p - 2p and 2p -2s overlapping has the order :-

(A) s – s > p – p > s – p

(B) s – s > p – s > p – p

(C) p – p > s – p > s – s

(D) p – p > s – s > p – s

Q43. π bond is formed :-

(A) By overlapping of hybridised orbitals

(B) Overlapping of s - s orbitals

(C) Head on overlapping of p -p orbitals

(D) By p - p collateral overlapping

Q44. Fluorine molecule is formed by :

(A) the axial p–p orbital overlap

(B) the sideways p–p orbital overlap

(C) the s–s orbital overlap

(D) the s–p orbital overlap

Q45. Consider the following statements :

I. A sigma (σ) bond is formed when two s-orbitals overlap

II. A pi(π) bond is formed when two p-orbitals axially overlap

III. A σ - bond is weaker than π - bond

Which of the above statements is / are correct?

(A) I and II (B) II and III

(C) I alone (D) II alone

Q46. Number of σ and π bonds present in $CH_3 - CH = CH - C \equiv CH$ are ?

(A) 10 s , 3 p (B) 10 s,2p

(C) 9s, 2p (D) 8s, 3p

Q47. Which of the following statements regarding covalent bond is not true ?

(A) The electrons are shared between atoms

(B) The bond is always non-directional

(C) The strength of the bond depends upon the extent of overlapping

(D) The bond formed may or may not be polar

Hybridisation

Q48. Overlapping of 2 hybrid orbitals can lead to the formation of :

(A) None

(B) σ–bond

(C) π –bond

(D) (B) and (C) both

Q49. In the compound

$\overset{1}{CH_2}=\overset{2}{CH}-\overset{3}{CH_2}-\overset{4}{CH_2}-\overset{5}{C}\equiv\overset{6}{CH}$, the $C_2 - C_3$ bond is formed by the overlapping of :-

(A) $sp - sp^2$ (B) $sp^3 - sp^3$

(C) $sp - sp^3$ (D) $sp^2 - sp^3$

Q50. Which of the set of species have same hybridisation state but different shapes?

(A) NO_2^+, NO_2 , NO_2^-

(B) ClO_4^-, SF_4, XeF_4

(C) NH_4^+, H_3O^+, OF_2

(D) SO_4^{-2}, PO_4^{-3}, ClO_4^-

Q51. Which compound has four sigma bond and one lone pair ?

(A) NH_4^+ (B) ICl_4^-

(C) SF_4 (D) XeF_4

Q52. Descending order of electronegativity of sp^3, sp^2 & sp hybridised orbitals is ?

(A) sp^2, sp, sp^3 (B) sp^3, sp^2, sp

(C) sp, sp^2. sp^3 (D) sp, sp^3, sp^2

Q53. In the protonation of H_2O, change occurs in

(A) Hybridisation state of oxygen

(B) Shape of molecule

(C) Hybridisation and shape both

(D) None

Q54. The d- orbitals involved in sp^3d hybridisation is:-

(A) $d_{x^2-y^2}$ (B) d_{z^2}

(C) d_{xy} (D) d_{xz}

Q55. In which of the hybridisation state different quantum No. are used:-

(A) sp^3 d^3 (B) sp^3d

(C) d^2sp^3 (D) sp

Q56. Which of the following species are expected to be planar:-

(a) NH_3 (b) NH_3^{2+}

(c) CH_3^+ (d) PCl_3

The correct answer is:-

(A) b and c (B) c and d

(C) b and d (D) a and d

Q57. Which order of decreasing bond angle is correct:-

(A) $CCl_4 > BF_3 > NO_2^+$

(B) $NH_3 > NCl_3 > NBr_3$

(C) $Br_2O > Cl_2O > OF_2$

(D) $PCl_3 > PBr_3 > PI_3$

Q58. A molecule is square planar with no lone pair. What type of hybridization is associated with it ?

(A) sp^3d (B) sp^3d^2

(C) dsp^3 (D) dsp^2

Q59. Hybridisation state of I in ICl_2^+ is
(A) dsp^2 (B) sp
(C) sp^3 (D) sp^2

Q60. The hybrid states of central atom in diaborane, diamond and graphite are respectively :-
(A) sp^2, sp^3, sp^2 (B) sp^3, sp^3, sp^2
(C) sp^3, sp^3, sp^3 (D) sp, sp^2, sp^3

Q61. The change in the state of hybridization of asterisked carbon in the following reaction is:-

$$CH_3\overset{*}{C}N \xrightarrow{H_2O} CH_3\overset{*}{C}ONH_2$$

(A) sp^3 to sp^2 (B) sp^2 to sp
(C) sp to sp^2 (D) sp^2 to sp^3

Bond Parameters

Q62. Which of the following has been arranged in order of decreasing bond length ?
(A) P–O > Cl–O > S–O
(B) P–O > S–O > Cl–O
(C) S–O > Cl–O > P–O
(D) Cl–O > S–O > P–O

Q63. When the hybridization state of carbon atom changes from sp^3, sp^2 and sp, the angle between the hybridized orbitals?
(A) decrease considerably
(B) increase progressively
(C) decrease gradually
(D) all of these

Q64. Which is correct statement ?
As the s-character of a hybrid orbital decreases
(I) The bond angle decreases
(II) The bond strength increases
(III) The bond length increases
(IV) Size of orbital increases
(A) (I), (III) and (IV)
(B) (II), (III) and (IV)
(C) (I) and (II)
(D) All are correct

Q65. The order of Bond Angle in CH_4 (A) and CCl_4 (B) is?
(A) A > B (B) A < B
(C) A = B (D) None of these

Bent's Rule and Drago's

Q66. Among the following, the correct statement is ?
(A) Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional
(B) Between NH_3 and PH_3, PH_3 is a better electron donor because the lone pair of electrons occupies sp^3orbital and is more directional
(C) Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional
(D) Between NH_3 and PH_3, PH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

Q67. In BClBr I molecule the maximum % s-character provided from the central atom is in bond ?
(A) B – I
(B) B – Cl
(C) B – Br
(D) Can not predict

Q68. T-shape molecules are
(A) ClF_3 (B) ICl_3
(C) BrF_3 (D) All of these

Q69. Which of the following two are isostructural :-
(A) XeF_2, IF_2^-
(B) NH_3,BF_3
(C) CO_3^{2-}, SO_3^{2-}
(D) PCl_5, ICl_5

Q70. The xenon compound(s) that are iso-structural with IBr_2^- and BrO_3^- respectively are :

(A) Linear XeF_2 and pyramidal XeO_3

(B) Bent XeF_2 and pyramidal XeO_3

(C) Bent XeF_2 and planar XeO_3

(D) Linear XeF_2 and tetrahedral XeO_3

Back Bonding

Q71. Type of back bonding in $(SiH_3)_2O$ is :

(A) $p\pi - d\pi$ (B) $p\pi - p\pi$

(C) $d\pi - d\pi$ (D) None of these

Q72. Boron forms BX_3 type of halides. The correct increasing order of Lewis-acid strength of these halides is -

(A) $BF_3 > BCl_3 > BBr_3 > BI_3$

(B) $BI_3 > BBr_3 > BCl_3 > BF_3$

(C) $BF_3 > BI_3 > BCl_3 > BBr_3$

(D) $BF_3 > BCl_3 > BI_3 > BBr_3$

Q73. Select species which is planar at nitrogen :

(A) $(CH_3)_3N$ (B) $(SiH_3)_3N$

(C) NF_3 (D) NH_3

Q74. Which of the following compounds has a 3-centred, 2 electron bond :–

(A) BF_3 (B) NH_3

(C) B_2H_6 (D) CO_2

Molecular Orbital Theory

Q75. Which of the following option w.r.t. increasing bond order is correct ?

(A) $NO < C_2 < O_2^- < He_2^+$

(B) $C_2 < NO < He_2^+ < O_2^-$

(C) $He_2^+ < O_2^- < NO < C_2$

(D) $He_2^+ < O_2^- < C_2 < NO$

Q76. The ion that is isoelectronic with CO and has same bond order is ?

(A) CN^- (B) O_2^+

(C) O_2^- (D) N_2^+

Q77. In which of the species, bond order increases on removing one electron ?

(a) NO (b) CN^-

(c) O_2 (d) CO

Correct answer is -

(A) b and d (B) a and c

(C) b, d and c (D) b and c

Q78. During the formation of a molecular orbital from atomic orbitals, probability of electron density is?

(A) minimum in the nodal plane

(B) maximum in the nodal plane

(C) zero at the nodal plane

(D) zero on the surface of the lobe

Q79. In the following which of the two are paramagnetic?

(A) N_2 (B) CO

(C) B_2 (D) NO_2

Correct answer is :-

(A) A and C (B) B and C

(C) C and D (D) B and D

Q80. Increasing order of bond length in NO, NO^+ and NO^- is ?

(A) $NO > NO^- > NO^+$

(B) $NO^+ < NO < NO^-$

(C) $NO < NO^+ < NO^-$

(D) $NO < NO^+ = NO^-$

Q81. Which of the following has fractional bond order ?

(A) O_2^{2+} (B) O_2^{2-}

(C) F_2^{2-} (D) H_2^+

Q82. On the basis of molecular orbital theory which molecules do not exist?

(A) H_2 (B) He_2

(C) HeH (D) Li_2

Q83. The bond order of CO molecule on the basis of molecular orbital theory is?

(A) Zero (B) 2

(C) 3 (D) 1

Q84. Which of the following group of molecules have 2.5 bond order ?

(A) N_2^{-2}, O_2^{-2}, CO

(B) N_2^{+}, O_2^{+}, NO

(C) C_2^{-2}, BN, O_2

(D) CN^{-}, NO^{+}, O_2^{+2}

Q85. Which of the following species will have the minimum bond energy ?

(A) N_2 (B) N_2^{-}

(C) N_2^{+} (D) N_2^{-2}

Q86. N_2 and O_2 are converted into monoanions, N_2^{-} and O_2^{-} respectively. Which of the following statements is wrong ?

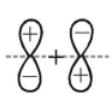
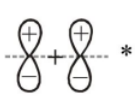
(A) In N_2^{-}, N-N bond weakens

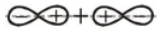
(B) In O_2^{-}, O-O bond order increases

(C) In O_2^{-}, O-O bond order decreases

(D) N_2^{-} becomes paramagnetic

Q87. Which of the following overlapping indicates formation of bonding molecular orbitals having gerade symmetry ?

(A)  (B) 

(C)  (D) All of these

Q88. In a homonuclear molecule which of the following set of orbitals are degenerate ?

(A) s_{2s} and s_{1s}

(B) p_{2px} and p_{2py}

(C) p_{2px} and s_{2pz}

(D) s_{2pz} and p^*_{2px}

Dipole Moment/Polarity

Q89. Which statement is correct ?

(A) All the compounds having polar bonds, have dipole moment

(B) SO_2 is non-polar

(C) H_2O molecule is non polar, having polar bonds

(D) PH_3 is polar molecule having non polar bonds

Q90. Which of the following molecule has zero dipole moment ?

(A) BF_3 (B) CH_2Cl_2

(C) NF_3 (D) SO_2

Q91. Which of the following species are polar ?

(a) C_6H_6 (b) XeF_2

(c) SO_2 (d) SF_4

(e) SF_6

Correct answer is :-

(A) (b) and (d)

(B) (a), (b) and (e)

(C) (a) and (e)

(D) (c) and (d)

Q92. Bond length of HCl is 1.275 Å (Charge = 4.8×10^{-10} e.s.u.) if μ = 1.02 D, then HCl is :

(A) 100% ionic

(B) 83% covalent

(C) 50% covalent

(D) 40% ionic

Q93. BeF_2 has zero dipole moment where as H_2O has a dipole moment because :-

(A) Water is linear

(B) H_2O is bent

(C) F is more electronegative than O

(D) Hydrogen bonding is present in H_2O

Q94. Which of the following species are symmetrical :-

(a) XeF_4 (b) XeF_6

(c) SO_2 (d) NH_3^{+2}

Correct answer is:-

(A) a and b (B) b and c

(C) c and d (D) a and d

Q95. In the compounds CH_3OH, CH_4, CF_4, CO_2, which has maximum dipole moment:-

(A) CH_3OH

(B) CF_4

(C) CH_4

(D) CF_4 and CO_2 have equally more

Q96. The dipole moment of NH_3 is:-

(A) Less than dipole moment of NCl_3

(B) Higher than dipole moment of NCl_3

(C) Equal to the diple moment of NCl_3

(D) None of these

Q97. PCl_5 is non polar because :-

(A) P – Cl bond is non-polar

(B) Its dipole moment is zero

(C) P – Cl bond is polar

(D) P & Cl have equal electronegativity

Q98. Which contains both polar and non-polar bonds ?

(A) NH_4Cl (B) HCN

(C) H_2O_2 (D) CH_4

Q99. Which of the following molecule does not possess a permanent dipole moment :

(A) H_2S (B) SO_2

(C) SO_3^{2-} (D) CS_2

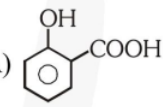
Hydrogen Bond and Weak Bonds

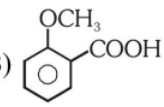
Q100. The hydrogen bond is strongest in :-

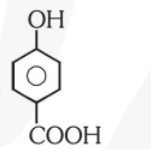
(A) O – H - - - S (B) S – H - - - O

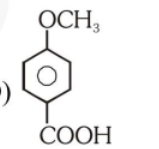
(C) F – H - - - F (D) O – H - - - O

Q101. Which of the following has strongest intra molecular hydrogen bonding ?

(A) 

(B) 

(C) 

(D) 

Q102. The correct order of volatility is:-

(A) $NH_3 < H_2O$

(B) p– nitro phenol < o– nitro phenol

(C) $CH_3OH > CH_3 – O – CH_3$

(D) HF > HCl

Q103. Pure phosphoric acid is very viscous. The reason is?

(A) It is a strong acid

(B) It is tribasic acid

(C) It is hygroscopic

(D) It has PO_4^{3-} groups which are bonded by many hydrogen bonds

Q104. Acetic acid exists as dimer in benzene due to?

(A) Condensation reaction

(B) Hydrogen bonding

(C) Presence of carboxyl group

(D) None of the above

Q105. Density of ice is less than that of water because of ?

(A) presence of hydrogen bonding

(B) crystal modification of ice

(C) open porous structure of ice due to hydrogen bonding

(D) different physical states of these

Q106. In dry ice the bond present between two molecules is ?

(A) Ionic bond (B) Covalent bond

(C) Hydrogen bond (D) Vander Waal

Exercise - 1A *Concept Builder*

MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Most ionic compounds have:

(A) high melting points and low boiling points

(B) high melting points and nondirectional bonds

(C) high solubilities in polar solvents and low solubilities in nonpolar solvents

(D) three-dimentional network structures, and are good conductors of electricity in the molten state

Q2. Which of the following is/are correct ?

(A) An ionic solute will dissolve in water if hydration energy of the ions is greater than the lattice energy

(B) Greater the size of the ions greater is the lattice energy

(C) Correct order of solubilities of the alkali metal fluorides is LiF < NaF < KF

(D) The correct order of the solubilities of the alkali metal nitrate is $LiNO_3 > NaNO_3 > KNO_3$

Q3. Polarization may be called the distortion of the shape of an anion by an adjacently placed cation. Which of the following statements is/are incorrect ?

(A) Minimum polarization is brought about by a cation of low radius

(B) A large cation is likely to bring about a large degree of polarization

(C) Maximum polarization is brought about by a cation of high charge

(D) A small anion is likely to undergo a large degree of polarization

Q4. Which of the following statements is/are true for BaO and MgO ?

(A) BaO is more ionic than MgO.

(B) MgO is more ionic than BaO.

(C) BaO has a higher melting point than MgO.

(D) MgO has a higher melting point than BaO.

Q5. Choose the correct order for the given properties.

(A) $MgSO_4 < SrSO_4 < BaSO_4$: Thermal stability order

(B) $BeC_2O_4 < CaC_2O_4 < BaC_2O_4$: Solubility in water

(C) LiCl > NaCl > KCl : Melting point order

(D) $BeF_2 > CaF_2 > SrF_2$: Covalent character order

Q6. Choose the correct order for the given properties.

(A) $NaF < MgF_2 < AlF_3$: covalent character order.

(B) $NaF < MgF_2 < AlF_3$: melting point order

(C) $NaF < MgF_2 < AlF_3$: lattice energy order
(D) $NaF > MgF_2 > AlF_3$: order of polarising power of cation.

Q7. Which of the following compounds contain/s both ionic and covalent bonds ?
(A) NH_4Cl (B) KCN
(C) $CuSO_4.5H_2O$ (D) NaOH

Q8. The octet rule is not obeyed in ?
(A) CO_2 (B) BCl_3
(C) PCl_5 (D) SiF_4

Q9. Which of the following species contain coordinate bond ?
(A) $AlCl_3$ (B) CO
(C) NOF_3 (D) N_3^-

Q10. The linear structure is assumed by ?
(A) $SnCl_2$ (B) NCO^-
(C) CS_2 (D) NO_2^+

Q11. Shape of NH_3 is very similar to ?
(A) SeO_3^{2-} (B) CH_3^-
(C) BH_3 (D) CH_3^+

Q12. Which of the following species are isoelectronic with CO_2 ?
(A) N_3^- (B) $(CNO)^-$
(C) $(NCN)^{2-}$ (D) NO_2^-

Q13. A π-bond may form between two p_x orbitals containing one unpaired electron each when they approach each other appropriately along ?
(A) x-axis (B) y-axis
(C) z-axis (D) any direction

Q14. Which combination of the compounds and their geometry/ shape are correct ?
(A) $HgCl_2$ - linear
(B) XeF_5^+ - square pyramidal
(C) ClF_3 - T-shaped
(D) ICl_4^-. square-planar

Q15. Select the correct statement for following molecules :
(I) $PF_2(CH_3)_3$; **(II)** $PF_2(CF_3)_3$
(A) Both have trigonal bipyramidal structure with respect to P.
(B) P-F bond length is longer in $PF_2(CH_3)_3$ than in $PF_2(CF_3)_3$
(C) F-atoms occupy axial position in both
(D) None of these

Q16. Molecules in which bond angle is changed due to back bonding w.r.t B/O/N ?
(A) H_3BO_3 (B) $B(OMe)_3$
(C) BF_3 (D) $N(SiH_3)_3$

Q17. $3d_\pi$-$2p_\pi$ type back bonding is observed in ?
(A) $N(SiH_3)_3$ (B) $\bar{C}Cl_3$
(C) $S(CH_3)_2$ (D) BCl_3

Q18. Three centre - two electron bonds exist in :
(A) B_2H_6 (B) $Al_2(CH_3)_6$
(C) $BeH_2(s)$ (D) $BeCl_2(s)$

Q19. Select correct about NO_2 :
(A) It is odd electron species
(B) N–O bond order = 1.5
(C) Paramagnatic species
(D) Isoelectronic with CO_2

Q20. Which of the following statements are **CORRECT** ?
(A) The free electron of ClO_3 molecule is present in d-orbital of Cl-atom
(B) The free electron of $\dot{C}F_3$ is present in sp^3 hybrid orbital
(C) NO is polar
(D) The free electron of ClO_2 molecule is present in d-orbital of Cl-atom

Q21. Which of the following do not exist ?
(A) SH_6 (B) HFO_4
(C) FeI_3 (D) $HClO_3$

Q22. Which of the following have identical bond order ?

(A) O_2^{2+} (B) NO^+

(C) CN^- (D) CN^+

Q23. Given the species : N_2, CO, CN^- and NO^+. Which of the following statements are true for these:

(A) All species are paramagnetic

(B) The species are isoelectronic

(C) All the species have dipole moment

(D) All the species are linear

Q24. Which of the following is/are correct ?

(A) During N^+_2 formation, one electron is removed from the bonding molecular orbital

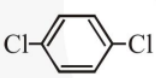
(B) During O^+_2 formation, one electron is removed from the antibonding molecular orbital

(C) During O^-_2 formation, one electron is added to the antibonding molecular orbital

(D) During CN^- formation, one electron is added to the bonding molecular orbital

Q25. Which has (have) zero value of dipole moment?

(A) XeF_4 (B) $CHCl_3$

(C) CO_2 (D) 

ASSERTION AND REASON

Q26. **Statement-1 :** C_3O_2 has linear structure.

Statement-2 : Each C atom in C_3O_2 is sp-hybridised.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q27. **Statement-1 :** Boron does not show univalent nature but unipositive nature of thallium is quite stable.

Statement-2 : Inert pair effect predominates in thallium.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q28. **Statement-1 :** Among alkali metal cations, Li^+(aq.) has highest electrical conductance.

Statement-2 : Li^+(aq.) is largest alkali metal cation because of greater degree of hydration.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q29. **Statement-1:** H_2 molecule is more stable than He–H molecule.

Statement-2 : The antibonding electron in He–H molecule decreases the bond order and there by the stability.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 30 to 32)

Hydridisation helps to determine the shape of a molecule but shape of a molecule is mainly defined by VSEPR theory According to this theory l.p-l.p > l.p – b.p > b.p – b.p repulsion.

Q30. In which of the following pairs hybridisation of the central atom is not same?

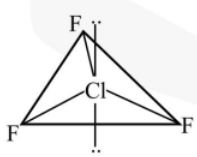
(A) ClF_3, ClF_3O,

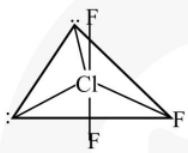
(B) ClF_3O, ClF_3O_2

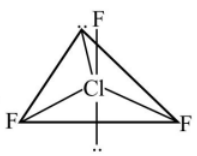
(C) $[ClF_2O]^+$ $[ClF_4O]^-$

(D) $[ClF_4O]^-$ $[XeOF_4]$

Q31. Correct structure of ClF_3 is ?

(A) 

(B) 

(C) 

(D) All of the above

Q32. Which one is the only V-shaped molecule or ion ?

(A) I_3^- (B) N_3^- (C) S_3^{2-} (D) C_3^{4-}

Paragraph # 2 (Q. 33 to 35)

Molecular orbital theory is based on linear combination of atomic orbitals (LCAO). According to LCAO when respective atomic orbitals of the atoms interact, they undergo constructive and destructive interference giving two types of molecular orbital i.e. bonding and antibonding molecular orbitals respectively.

Q33. Which of the following species is paramagnetic ?

(A) NO^- (B) O_2^{2-}

(C) CN^- (D) CO

Q34. Bond order of Be_2 is ?

(A) 1 (B) 2

(C) 3 (D) 0

Q35. Number of anti bonding electrons in N_2 is?

(A) 4 (B) 10

(C) 12 (D) 14

Paragraph # 3 (Q. 36 to 38)

Bond angles can be explained by the electro negativity or by the size. Species with larger difference in electronegativity values between central and outer atoms have smaller bond angles. As the size of outer atom increases ,the angle increases. The size of central atom can also be used to determine the bond angles in the series.

Q36. Consider the following molecules

H_2O	H_2S	H_2Se	H_2Te
(I)	(II)	(III)	(IV)

the increasing order of bond angles is -

(A) I < II < III < IV (B) IV < III < II < I

(C) I < II < IV < III (D) II < IV < III < I

Q37. Which has the smallest bond angle (XSX) in the given molecules ?

(A) OSF_2 (B) $OSCl_2^-$

(C) $OSBr_2$ (D) OSI_2

Q38. The bond angles of the iodides are as follows.

PI_3	AsI_3	SbI_3
102°	100.2°	99°

The bond angle is maximum in PI_3 because-

(A) due to small size of phosphorus

(B) due to more bp-bp repulsion in PI_3

(C) due to less electronegativity of P

(D) All of these

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q39. Match list I with list II and select the correct answer:

List–I (Molecules)	**List–II (Hybridisation)**
(P) SF_6	(1) sp^3d^2
(Q) BF_3	(2) sp^3d
(R) PCl_3	(3) sp^3
(S) ClF_3	(4) sp^2

Code :

	P	**Q**	**R**	**S**
(A)	3	2	1	4
(B)	1	4	3	2
(C)	4	3	2	1
(D)	1	2	3	4

Q40. Match list I with list II and select the correct answer:

List–I (Species)	**List–II (Number of lone pairs on molecule)**
(P) XeF_2	(1) 14
(Q) XeF_4	(2) 9
(R) XeF_6	(3) 18
(S) SF_6	(4) 19

Code :

	P	**Q**	**R**	**S**
(A)	2	1	4	3
(B)	3	2	1	4
(C)	4	3	2	1
(D)	1	2	3	4

Q41. Match list I with list II and select the correct answer:

List–I (Process)	**List–II (Operating intraction involved)**
(P) Clathrate compound of Xe in ice	(1) Ion - Dipole
(Q) Liquation of Xe gas	(2) Dipole - Dipole
(R) Liquation of HCl gas	(3) Dipole - Induced dipole
(S) Hydration of Na^+	(4) London forces

Code :

	P	Q	R	S
(A)	1	3	4	2
(B)	3	4	1	2
(C)	3	4	2	1
(D)	1	3	2	4

Q42. Match list I with list II and select the correct answer:

List I (species)

(P) NO^{+}_{2}

(Q) NO_2

(R) NO^{-}_{2}

(S) NO^{-}_{3}

List II (O-N-O angle)

(1) 180°

(2) 134°

(3) 120°

(4) 115°

(5) 109°

Code :

	P	Q	R	S
(A)	5	4	3	2
(B)	5	2	4	3
(C)	1	2	4	3
(D)	1	4	3	2

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q43. How many of the following are paramagnetic-

C_2, B_2, O_2^{-2}, BN, Cl_2^{+}, NO^{+}.

Q44. Find the number of molecules / ions which do not exist due to steric repulsion

PI_5(gas) , PCl_6^{-}, PI_6^{-} , PI^{+}_{4} , PBr_6^{-} , PH_5

Q45. The number of resonating structures for the azide ion, N_3^{-} are?

Q46. The observed dipole moment of H_2O = 1.85 D. The H – O bond distance is 0.94 Å and HOH bond angle is 105 degrees.

cos 52.5 degree = 0.609 ;

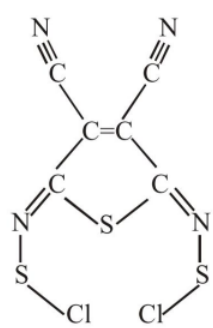
cos 105 degree = – 0.26

Determine the percentage fractional charges on each oxygen atom in terms of charge of electron in nearest possible integers.

Charge of electron = 4.8×10^{-10} e. s. u.

Q47. Write the number of FIF angles which are less than 90° in IF_5.

Q48. Write the maximum number of atoms having same hybridisation in 2, 5-di-N-chloridothioimino-3, 4 diagonothiophene as shown below :



Q49. Find the ratio of π-electrons in the C_2-molecule with that of B_2 molecule according to M.O.T.

Q50. The total number of bonding and antibonding electrons in O_2^+ are "....." and "....." respectively.

[If the answer is 14 and 7, then represent as 147]

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Which of the following statements is correct for carbon monoxide ?

(A) A double bond between carbon and oxygen

(B) 1 σ , 1 π and 1 coordinate bond between carbon and oxygen

(C) one lone pair of electrons only on oxygen atom

(D) 1 s and 2 p bonds between carbon and oxygen.

Q2. Identify isostructural pairs from (I) NF_3, (II) NO_3^-, (III) BF_3, (IV) H_3O^+, (V)HN_3

(A) I & II, III & IV

(B) I & V, II & III

(C) I & IV, II & III

(D) I & IV, III & V

Q3. Which of the following diagram / statement is correct for the hybrid orbitals.

(A) $s + p_y \longrightarrow z'$  z

(B) $s + p_y + p_z \longrightarrow sp^2$ orbitals lie in the xy plane

(C) $ns + np^3 +$ any $d^2 \longrightarrow$ results in octahedral geometry of sp^3d^2 hybrid orbitals.

(D) None of these

Q4. A : tetracyanomethane

B : Carbondioxide

C : Benzene

D : 1, 3-buta-di-ene

Ratio of σ and π bonds is in order ?

(A) A = B < C < D

(B) A = B < D < C

(C) A = B = C = D

(D) C < D < A < B

Q5. The hybridisation of the central atom will change when ?

(A) NH_3 combines with H^+

(B) H_3BO_3 combines with OH^-

(C) NH_3 form NH_2^-

(D) H_2O combines with H^+

Q6. C–H bond distance is the longest in ?

(A) C_2H_2 (B) C_2H_4

(C) C_2H_6 (D) $C_2H_2Br_2$

Q7. The correct order of increasing X–O–X bond angle is (X = H, F or Cl) ?

(A) $H_2O > Cl_2O > F_2O$

(B) $Cl_2O > H_2O > F_2O$

(C) $F_2O > Cl_2O > H_2O$

(D) $F_2O > H_2O > Cl_2O$

Q8. In which of the following case C—C bond length will be highest ?

(A) $CF_3 - CF_3$ (B) $F_2CH - CHF_2$

(C) $FCH_2 - CH_2F$ (D) $CH_3–CF_3$

Q9. Find out the % p-character in the orbital occupied by lone pairs in H_2O.
HOH = 104°5 and cos (104.5) = –0.25
(A) 80 % (B) 20 %
(C) 70 % (D) 75 %

Q10. Type of back bonding in $(SiH_3)_2O$ is ?
(A) pπ – dπ (B) pπ – pπ
(C) dπ – dπ (D) None of these

Q11. In which of the following cases, the strength of back bonding is maximum?
(A) PF_3 (B) BF_3
(C) $(H_3Si)_3N$ (D) $(SiH_3)_2O$

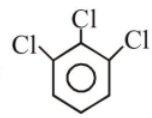
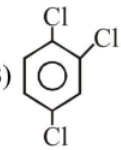
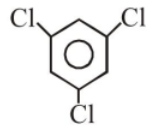
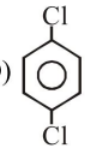
Q12. The type of overlap in the bridge bond existing in $Al_2(CH_3)_6$ is ?
(A) sp^3–sp^3d–sp^3 (B) sp^3–sp^2–sp^3
(C) sp^3–s–sp^3 (D) sp^3–sp^3–sp^3

Q13. Which of the following molecules do not exist?
(A) XeF_3^- (B) NCl_3
(C) NOF_3 (D) XeF_5^-

Q14. The bond energy order of He_2^+ and HeH^+ is-
(A) $He_2^+ > HeH^+$
(B) $HeH^+ = He_2^+$
(C) $HeH^+ > He_2^+$
(D) Can't be predicted

Q15. Which of the following boiling point order is correct ?
(A) $H_2 > T_2 > D_2$ (B) $H_2 < T_2 < D_2$
(C) $H_2 > He > D_2$ (D) $H_2 < D_2 < T_2$

Q16. If a molecule MX_3 has zero dipole moment, the sigma bonding orbitals used by M (atm. no. < 21) are ?
(A) pure p (B) sp hybrid
(C) sp^2 hybrid (D) sp^3 hybrid

Q17. Which has maximum dipole moment?
(A) 
(B) 
(C) 
(D) 

Q18. The bond angles of NH_3, NH_4^+ and NH_2^- are in the order :
(A) $NH_2^- > NH_3 > NH_4^+$
(B) $NH_4^+ > NH_3 > NH_2^-$
(C) $NH_3 > NH_2^- > NH_4^+$
(D) $NH_3 > NH_4^+ > NH_2^-$

Q19. Which of the following has been arranged in order of decreasing dipole moment ?
(A) $CH_3Cl > CH_3F > CH_3Br > CH_3I$
(B) $CH_3F > CH_3Cl > CH_3Br > CH_3I$
(C) $CH_3Cl > CH_3Br > CH_3I > CH_3F$
(D) $CH_3F > CH_3Cl > CH_3I > CH_3Br$

Q20. Weakest bond is formed by the orbital overlapping of ?
(A) $sp^2 – s$ (B) $sp^3 – p$
(C) s – s (D) p – p co-axial

Q21. Which contains both covalent and ionic bonds ?
(A) NH_4Cl (B) HCN
(C) H_2O_2 (D) CH_4

Q22. Which of the following is correct for CO_2 ?
(A) nonpolar with polar bonds
(B) polar with nonpolar bonds
(C) polar with polar bonds
(D) all of the above are wrong

Q23. Which of the following molecules are polar ?

(A) ClO_2 (B) SO_2

(C) NO_2 (D) All of the given

Q24. In a change from $PCl_3 \rightarrow PCl_5$, the hybrid state of P changes from :-

(A) sp^2 to sp^3 (B) sp^3 to sp^2

(C) sp^3 to sp^3d (D) sp^3 to dsp^2

Q25. Which of the following is an example of super octet molecule ?

(A) ClF_3 (B) PCl_5

(C) IF_7 (D) All the three

Q26. The linear structure is not assumed by ?

(A) $SnCl_4$ (B) NCO^-

(C) CS_2 (D) NO^+_2

Q27. Among the following molecules which one is planar ?

(A) BCl_3 (B) SO_2Cl_2

(C) NH_3 (D) NF_3

Q28. Maximum bond energy will be shown by the species?

(A) O_2^+ (B) O_2

(C) O_2^- (D) O_2^{-2}

Q29. The nature of intermolecular forces among benzene (C_6H_6) molecules is ?

(A) Dipole-dipole attraction

(B) London dispersion forces

(C) Ion-dipole attraction

(D) Hydrogen bonding

Q30. The compound formed by which by of the following pair of ions will have lowest melting point ?

(A) Na^+ and Cl^- (B) Mg^{2+} and Cl^-

(C) Al^{3+} and Cl^- (D) Sn^{4+} and Cl^-

Q31. Which of the following can not be explained by VBT ?

(A) Overlapping

(B) Bond formation

(C) Paramagnetic nature of oxygen

(D) Shapes of molecules.

Q32. In an antibonding molecular orbital, electron density is minimum ?

(A) Around one atom of the molecule

(B) Between the two nuclei of the molecule

(C) At the region away from the nuclei of the molecule

(D) at no place

Q33. Which of the following match is not correct?

(A) ICl_2^- – Linear ion

(B) ICl_4^- – Square planar ion

(C) XeF_2 – Linear molecule

(D) SO_4^{2-} - Trigonal planar ion

Q34. Which of the following statements is incorrect in the context of the B–F bond in BF_3 ?

(A) All the three B–F bond lengths are equal and each of them is shorter than the sum of the covalent radii of boron and fluorine.

(B) The bond energy of the B–F bond is very low, lower than for any other single bond.

(C) The unusual shortness and strength of the B–F bond may be explained by a $p\pi - p\pi$ interaction between boron and fluorine atoms.

(D) Back bonding is possible for boron.

Q35. Which of the following is not possible due to back bonding?

(A) State of hybridisation may change

(B) Bond order increases

(C) Bond angle always increases

(D) Lewis acidic strength decreases

Q36. The correct order of increasing C—O bond length of CO, CO_3^{2-}, CO_2 is ?

(A) $CO_3^{2-} < CO_2 < CO$

(B) $CO_2 < CO_3^{2-} < CO$

(C) $CO < CO_3^{2-} < CO_2$

(D) $CO < CO_2 < CO_3^{2-}$

Q37. Resonance is shown by ?

(A) CO_3^{2-} (B) NO_3^-

(C) O_2 (D) Both (A) and (B)

Q38. Amongst CO_3^{2-}, AsO_3^{3-}, XeO_3 , ClO_3^- , BO_3^{3-} and SO_3^{2-} the non-planar species are?

(A) XeO_3 , ClO_3^- , SO_3^{2-}, AsO_3^{3-}

(B) AsO_3^{3-} , XeO_3 , CO_3^{-2}

(C) BO_3^{3-} , CO_3^{2-} , SO_3^{2-}

(D) AsO_3^{3-} , BO_3^{3-} , CO_3^{2-}

Q39. The bond angle in H_2O molecule is less than that of NH_3 molecule because :-

(A) The hybridisation of O in H_2O and N in NH_3 is different

(B) The atomic radii of N and O are different

(C) There is one lone pair of electrons on O and two lone pairs of electrons on N

(D) There are two lone pairs of electrons on O and one lone pair of electrons on N.

Q40. According to molecular orbital theory which of the following statement about the magnetic character and bond order is correct regarding O_2^+.

(A) Paramagnetic and Bond order < O_2

(B) Paramagnetic and Bond order > O_2

(C) Diamagnetic and Bond order < O_2

(D) Diamagnetic and Bond order > O_2

Q41. What is the percentage of s- character in the orbital occupied by lone pair of NH_3.

It is given that : cos 107° = –0.292

(A) 22.6% (B) 32.2%

(C) 25% (D) None of these

Q42. Select the correct order of solubility (in water) from the following :

(A) $SrSO_4 < CaSO_4 < MgSO_4 < BeSO_4$

(B) NaF < KF < RbF < CsF

(C) $Ba(OH)_2 > Sr(OH)_2 > Ca(OH)_2 > Mg(OH)_2$

(D) All of these

Q43. The dissolution of ionic compounds involves :

(A) Evolution of heat

(B) Weakening of attractive forces

(C) Dissociation into ions

(D) All of these

Q44. Least melting point is shown by the compound ?

(A) $PbCl_2$ (B) $SnCl_4$

(C)NaCl (D)$AlCl_3$

Q45. Which of the following carbonate will not decompose on heating :-

(A) $BaCO_3$ (B) $ZnCO_3$

(C) Na_2CO_3 (D) Li_2CO_3

Q46. Which of the following sets of characteristics leads to the increase in solubility of ionic substances?

(A) High dipole moment, strong attraction by an ion towards solvent and large solvation energy

(B) Low dipole moment, weak attraction by an ion towards solvent and high solvation energy

(C) High dipole moment, strong attraction by an ion towards solvent and low solvation energy

(D) High dipole moment, weak attraction by an ion towards solvent and large solvation energy

MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following species contain coordinate bond ?

(A) $AlCl_3$ (B) CO

(C) NOF_3 (D) N_3^-

Q2. Shape of NH_3 is very similar to ?

(A) SeO_3^{2-} (B) CH_3^-

(C) BH_3 (D) CH_3^+

Q3. The statements that are true among the following are ?

(A) The bond order in CO molecule is 3

(B) The order of stability is $O_2^{2-} > O_2^- > O_2 > O_2^+$

(C) KHF_2, $KHCl_2$ and $KHBr_2$ are known but KHI_2 is not

(D) NO is paramagnetic while NO^+ is diamagnetic

Q4. CO_3^{2-} anion has which of the following characteristics ?

(A) Bonds of equal length

(B) sp2 hybridization of C atom

(C) Resonance stabilization

(D) Same bond angles

Q5. The type of bonds present in PH_4I are ?

(A) Ionic

(B) Covalent

(C) Co-ordinate

(D) H–bonds

Q6. Most ionic compounds have ?

(A) High melting points and low boiling points.

(B) High melting points and nondirectional bonds.

(C) High solubilities in polar solvents and low solubilities in nonpolar solvents.

(D) Three-dimensional network structures, and are good conductors of electricity in the molten state.

Q7. Which statement is correct about hybridization ?

(A) In hybridzation orbitals take part

(B) In hybridization electrons take part

(C) In hybridzation fully filled , half or empty orbitals can take part

(D) Hybridzed orbitals contain bond pair electrons only

Q8. Which of the following is/are correct for dipole moment ?

(A) Lone pair of elements present on central atom can give rise to dipole moment

(B) Dipole moment is a vector quantity

(C) CO_2 molecule has dipole moment

(D) Difference in electronegativities of combining atom can lead to dipole moment

Q9. Correct order of solubility in water of the compound is ?

(A) $Na_2SO_4 < K_2SO_4 < Rb_2SO_4$

(B) $Rb_2SO_4 < K_2SO_4 < Na_2SO_4$

(C) CsF < CsCl < CsBr

(D) CsBr < CsCl < CsF

Q10. Which of the following molecule are polar and sp^3d^2 hybridized ?

(A) IF_5 (B) XeF_4

(C)$XeOF_4$ (D) ICl_4

Q11. Which is/are true about VSEPR theory ?

(A) Lone pair-lone pair repulsion is maximum.

(B) Lone pair and double bond occupy equitorial position in trigonal bipyramidal structure.

(C) More electronegative atoms occupies axial position in trigonal bipyramidal structure.

(D) Bigger atoms occupy axial positions in trigonal bipyramidal structure.

Q12. Choose the correct statements :-

(A) d_{P-F}(axial) > d_{P-F}(equitorial) in PF_3Cl_2

(B) $d_{P-F} > d_{P-Cl}$ in PF_2Cl_3

(C) $d_{P-Cl}(PF_2Cl_3) > d_{P-Cl}(PF_3Cl_2)$

(D) All d_{P-Cl} (in PF_2Cl_3) are identical while all d_{P-F}(in PF_3Cl_2) are not identical

Q13. London force works in ?

(A) Polar molecule

(B) Non-polar molecule

(C) All polar and non-polar molecule

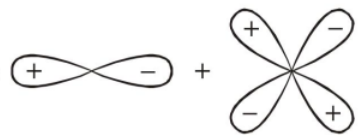
(D) None of these

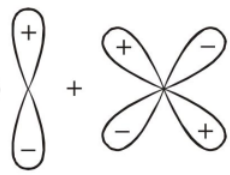
Q14. In which of the following process the electron is only added to bonding molecular orbital.

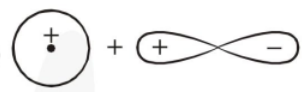
(A) $N^+_2 \rightarrow N^-_2$ (B) $O^+_2 \rightarrow O^-_2$

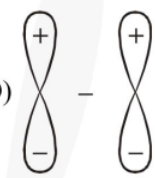
(C) $C_2 \rightarrow C_2^{2-}$ (D) $B_2 \rightarrow B_2^{2-}$

Q15. Correct set of overlapping of following atomic orbitals which gives bonding molecular orbital.

(A) 

(B) 

(C) 

(D) 

Q16. Which of the following statements is/are correct ?

(A) KF combines with HF to form KHF_2 which contains species K^+ and HF_2^-

(B) LiCl is a predominantly covalent molecule

(C) $CsBr_3$ contains Cs^+ and Br_3^- ions

(D) Sodium sulphate is soluble in water whereas Barium sulphate is sparingly soluble in water

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17. Match the column –

Column I	**Column II**
(A) CO_3^{2-}	(P) sp hybridization, linear
(B) $SiCl_4$	(Q) sp^3, tetrahedral
(C) C_3O_2	(R) sp^2, soft and conducting
(D) Graphite	(S) sp^2, trigonal planar

Q18.

Column I (pair of species)	**Column II (Identical properties in the pair of species)**
(A) CO_2 and N_2O	(P) Total number of bonds (s + p)
(B) C_6H_6 and $B_3N_3H_6$	(Q) Hybridization of central atom
(C) BF_3 and BCl_3	(R) Same shape of species
(D) I_3^- and I_3^+	(S) Total number of electrons

Q19.

Column I (Species)	**Column II (properties)**
(A) O_2^{2-}	(P) Isoelectric with N_2
(B) CO	(Q) Fractional bond order
(C) NO^+	(R) Paramagnetic
(D) He_2^+	(S) Diamagnetic

Q20.

Column-I	**Column-II**
(A) NH_4Cl	(P) Hydrogen bond
(B) $CuSO_4.5H_2O$	(Q) Co-ordinate bond
(C) HNC	(R) Ionic bond
(D) Liquid H_2O_2	(S) Covalent bond

Q.21

Column – I	**Column - II**
(A) $B_3N_3H_6$	(P) Planar geometry
(B) S_2Cl_2	(Q) Non-planar geometry
(C) B_2H_6	(R) No lone pair
(D) I_2Cl_6	(S) Non-polar molecule

Q.22

Column–I	**Column–II**
(A) $(CH_3)_2PF_3$	(P) Lone pair of electrons on central atom = 1
(B) SF_4	(Q) Central atom is sp^3d hybridised
(C) XeF_4	(R) Equal length of all central atom–fluorine bond
(D) BrF_3	(S) Total number of lone pair of electrons is more than 9

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # (Q.23 to Q.26)

PCl_5 in solid state exists as PCl_4^+ and PCl_6^-.

Also in some solvents it undergoes dissociation as $2PCl_5 \rightleftharpoons PCl_4^+ + PCl_6^-$

Q23. The geometry and hybridisation of PCl_5 is ?

(A) Trigonal bipyramid, sp^3d

(B) Tetrahedral, sp^3

(C) Octahedral, sp^3d^2

(D) None of these

Q24. The geometry and hybridiation of PCl_4^+ is ?

(A) Tetrahedral, sp^3

(B) Octahedral, sp^3d^2

(C) Trigonal pyramid, sp^3d

(D) See-saw, sp^3d

Q25. The geometry and hybridisation of PCl_6^- is?

(A) Octahedral, sp^3d^2

(B) Tetrahedral, sp^3

(C) Square planar bipyramid, sp^3d

(D) See-saw, sp^3

Q26. Find the number of molecules which are planar but not polar.

$H_2C=C=CH_2$, $B_3N_3H_6$, SiF_4 , C_3O_2 , NO_2

(A) 1 (B) 2

(C) 3 (D) 5

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
A	C	B	C	D	C	D	B	C	C
11	12	13	14	15	16	17	18	19	20
B	B	A	A	A	B	B	C	C	B
21	22	23	24	25	26	27	28	29	30
D	C	A	A	C	C	B	B	C	A
31	32	33	34	35	36	37	38	39	40
C	B	A	C	D	A	B	D	A	C
41	42	43	44	45	46	47	48	49	50
B	C	D	A	C	A	B	B	D	C
51	52	53	54	55	56	57	58	59	60
C	C	B	B	C	A	C	D	C	B
61	62	63	64	65	66	67	68	69	70
C	B	B	A	C	C	A	D	A	A
71	72	73	74	75	76	77	78	79	80
A	B	B	C	D	A	B	C	C	B
81	82	83	84	85	86	87	88	89	90
D	B	C	B	D	B	C	B	D	A
91	92	93	94	95	96	97	98	99	100
D	B	B	D	A	B	B	C	D	C
101	102	103	104	105	106				
A	B	D	B	C	D				

Ex-1A

1. B,C,D **2.** A,C,D **3.** A,B,D **4.** A,C **5.** A,D

6. A,B,C **7.** A,B,C,D **8.** B,C **9.** B,C,D **10.** B,C,D

11. A,B **12.** A,B,C **13.** B,C **14.** A,B,C,D **15.** A,B,C

16. D **17.** A,B **18.** A,B,C **19.** A,B,C **20.** B,C,D

21. A,B,C **22.** A,B,C **23.** B,D **24.** A,B,C,D **25.** A,C,D

26. A **27.** A **28.** D **29.** A **30.** C

31. B **32.** C **33.** A **34.** D **35.** A

36. B **37.** A **38.** B **39.** B **40.** A

41. C **42.** C **43.** 2 **44.** 3 **45.** 3

46. 67.2 **47.** 8 **48.** 6 **49.** 2 **50.** 105

Ex-2

1	2	3	4	5	6	7	8	9	10
B	C	D	A	B	C	B	C	C	A
11	12	13	14	15	16	17	18	19	20
B	D	A	C	D	C	A	B	A	C
21	22	23	24	25	26	27	28	29	30
A	A	D	C	D	A	A	A	B	D
31	32	33	34	35	36	37	38	39	40
C	B	D	B	C	D	D	A	D	B
41	42	43	44	45	46				
B	D	D	B	C	A				

Ex-2A

1. BCD **2.** AB **3.** AD **4.** ABCD **5.** ABC

6. BCD **7.** AC **8.** ABD **9.** BD **10.** AC

11. ABC **12.** ACD **13.** ABC **14.** CD **15.** BC

16. ABCD

17. A→ S, B→ Q, C→ P, D→ R

18. (A) →(P, Q,R, S), (B) → (P, Q, R, S), (C) → (P, Q, R), (D) → (P)

19. (A) →(S) (B) → (P, S), (C) → (P, S), (D) → (Q,R)

20. (A) → (Q, R, S) (B) → (P, Q, R, S) (C) → (Q, S) (D) → (P,S)]

21. (A) → (P, R, S), (B) →(Q), (C) → (Q, R, S), (D) → (P, S)

22. (A)→(Q) ; (B) - (P, Q,S) ; (C) - (R,S) (D) - (Q,S)

23. A **24.** A **25.** A **26.** B

-: Important Notes & Formulas :-

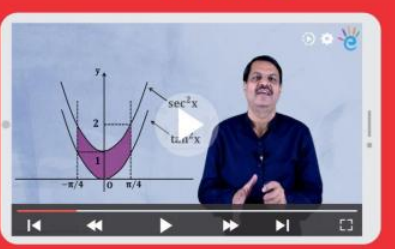
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